

Progress Test 3.A — Thermo-S1

3.A · Low-density gases: the historical laws

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

The product pV of a gas equals 2.0 bar L. Express it in joules.

$pV = 2.0 \times 10^5 \text{ Pa} \times 1.0 \times 10^{-3} \text{ m}^3$	$pV = 2.0 \times 10^2 \text{ J}$
(Pa m ³ = N m = J)	

Marking

①
0 1

Marking scheme

① 1 pt 200 J, with the unit.

Examiner's tip

pV has the dimension of an energy, which is why R is in J K⁻¹ mol⁻¹. Useful to remember:
1 bar L = 100 J.

Exercise 2. Boyle–Mariotte law

(5 marks)

State the Boyle–Mariotte law: the conditions under which it holds, and its formula. Deduce from it the isothermal compressibility χ_T of a low-density gas.

Boyle–Mariotte (1662): for a low-density gas at fixed temperature T (and fixed n), the product pV is constant, whatever the gas:	
$pV = \text{const.} \quad \text{or} \quad p_1 V_1 = p_2 V_2 \quad (T_1 = T_2)$	
So $V = C/p$ and $\left(\frac{\partial V}{\partial p}\right)_T = -\frac{C}{p^2} = -\frac{V}{p}$, hence	$\chi_T = -\frac{1}{V} \left(\frac{\partial V}{\partial p}\right)_T = \frac{1}{p}$ in
Pa⁻¹.	
The gas becomes less compressible as the pressure increases.	

Marking

①
0 1 2
②
0 1 2
③
0 1

Marking scheme

- ① **2 pts** Statement: fixed T and n , low density (1 pt); $pV = \text{const.}$ (1 pt).
② **2 pts** $(\partial V/\partial p)_T = -V/p$, derived.
③ **1 pt** $\chi_T = 1/p$, in Pa^{-1} .

Exercise 3. The three historical laws

(3 marks)

Charles's law holds at fixed:

- ☐ n and T ☒ n and p ☐ n and V ☐ p and V

Why: Tab. 4: Charles (1787), fixed n and p , free V and T : $V/T = \text{const.}$

Gay-Lussac's law states that, at fixed n and V :

- ☐ $pV = \text{const.}$ ☐ $V/T = \text{const.}$ ☒ $p/T = \text{const.}$ ☐ $pT = \text{const.}$

Why: equation (14) of the notes: $p_1/T_1 = p_2/T_2$ if $V_1 = V_2$.

For a low-density gas, the isobaric expansion coefficient is:

- ☒ $\alpha = 1/T$ ☐ $\alpha = 1/p$ ☐ $\alpha = T$ ☐ $\alpha = 0$

Why: $V/T = \text{const.}$ gives $(\partial V/\partial T)_p = V/T$, so $\alpha = 1/T$ (Box 15).

Exercise 4. A rising bubble and a heated balloon

(5 marks)

(a) An air bubble of volume V_1 is released at a depth where the pressure is p_1 . It rises to the surface, where the pressure is p_2 , at constant temperature. Find its volume V_2 at the surface. (b) A balloon of volume V at θ_1 is heated to θ_2 at constant pressure. Find its new volume V' .

$$\begin{array}{lll} \text{(a)} & V_1 = 2.0 \text{ cm}^3 & p_1 = 3.0 \text{ bar} & p_2 = 1.0 \text{ bar} \\ & V = 3.0 \text{ L} & \theta_1 = 27^\circ\text{C} & \theta_2 = 77^\circ\text{C} \end{array}$$

(a) Fixed n and T , so Boyle–Mariotte applies: $p_1 V_1 = p_2 V_2$, i.e.

$$V_2 = \frac{p_1 V_1}{p_2} = \frac{3.0}{1.0} \times 2.0 \text{ cm}^3 \quad V_2 = 6.0 \text{ cm}^3$$

(b) Fixed n and p , so Charles applies: $\frac{V}{T_1} = \frac{V'}{T_2}$, i.e. $V' = V \frac{T_2}{T_1}$, with **absolute** temperatures $T_1 = 300 \text{ K}$ and $T_2 = 350 \text{ K}$:

$$V' = 3.0 \text{ L} \times \frac{350}{300} \quad V' = 3.5 \text{ L}$$

Marking

①	0	1
②	0	1
③	0	1
④	0	1
⑤	0	1

Marking scheme

- ① 1 pt $V_2 = p_1 V_1 / p_2$ (literal).
- ② 1 pt $V_2 = 6.0 \text{ cm}^3$.
- ③ 1 pt $V' = V T_2 / T_1$ (literal).
- ④ 1 pt Temperatures converted into kelvins: 300 K and 350 K.
- ⑤ 1 pt $V' = 3.5 \text{ L}$.

Examiner's tip

The laws of Charles and Gay-Lussac need absolute temperatures. Using $^\circ\text{C}$ would give $V' = 3.0 \times 77/27 \approx 8.6 \text{ L}$, which is absurd.

Your score: / 14 → mark out of 20 = score $\times$ 20/14 = / 20